Operating Systems
Week 3
Lecture/Discussion 3

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Previous

- How OS services can be managed
- Role of an OS
- Process Control Block (Can be quiz)
- Examples of Real-Time systems

Today Discussion

- Role of Process Control Block
- Switching Processes

Role of the Process Control Block

The most important data structure in an OS

- It defines the state of the OS

Process Control Block requires protection

- A faulty routine could cause damage to the block destroying the OS's ability to manage the process
- Any design change to the block could affect many modules of the OS

Modes of Execution

Most processors support at least two modes of execution

User mode

- Less-privileged mode
- User programs typically execute in this mode

System mode

- More-privileged mode
- Kernel of the operating system

Process Creation

Once the OS decides to create a new process it:

- Assigns a unique process identifier
- Allocates space for the process
- Initializes process control block
- Sets up appropriate linkages
- Creates or expand other data structures

Switching Processes

Several design issues are raised regarding process switching

- What events trigger a process switch?
- We must distinguish between mode switching and process switching.
- What must the OS do to the various data structures under its control to achieve a process switch?

When to switch processes

A process switch may occur any time that the OS has gained control from the currently running process. Possible events giving OS control are:

Mechanism	Cause	Use
Interrupt	External to the execution of the current instruction	Reaction to an asynchronous external event
Trap	Associated with the execution of the current instruction	Handling of an error or an exception condition
Supervisor call	Explicit request	Call to an operating system function

Table 3.8 Mechanisms for Interrupting the Execution of a Process

Change of Process State …

The steps in a process switch are:

1. Save context of processor including program counter and other registers
2. Update the process control block of the process that is currently in the Running state
3. Move process control block to appropriate queue – ready; blocked; ready/suspend

Change of Process State cont…

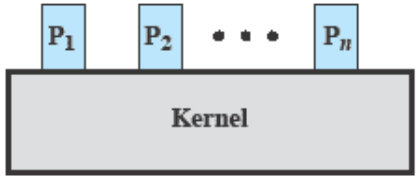
4. Select another process for execution
5. Update the process control block of the process selected
6. Update memory-management data structures
7. Restore context of the selected process

Is the OS a Process?

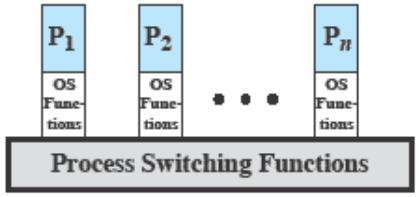
If the OS is just a collection of programs and if it is executed by the processor just like any other program, is the OS a process?
If so, how is it controlled?

- Who (what) controls it?

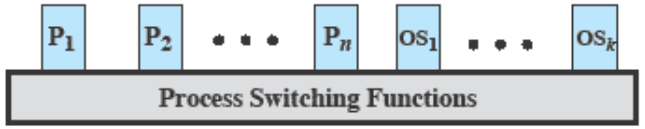
Execution of the Operating System



(a) Separate kernel



(b) OS functions execute within user processes

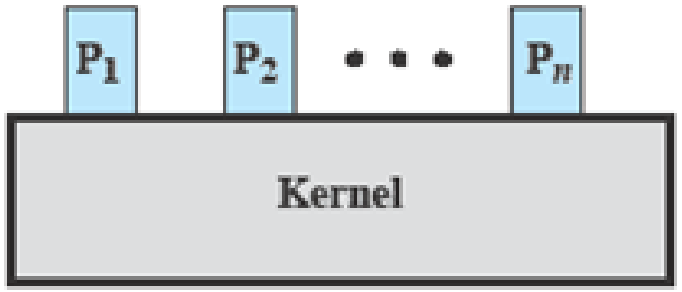


(c) OS functions execute as separate processes

Figure 3.15 Relationship Between Operating System and User Processes

Non-process Kernel

Execute kernel outside of any process
The concept of process is considered to apply only to user programs
- Operating system code is executed as a separate entity that operates in privileged mode

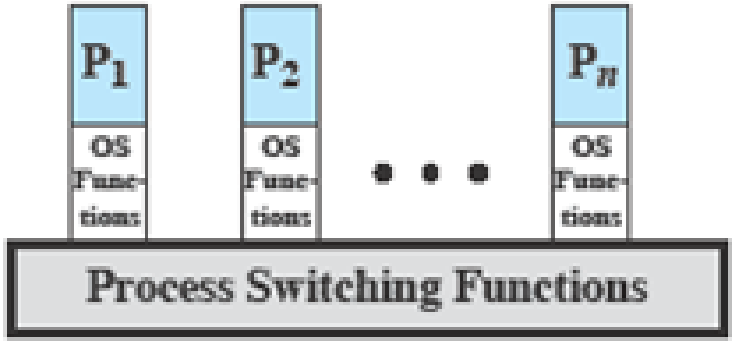


(a) Separate kernel

Execution *Within* User Processes

Execution Within User Processes

- Operating system software within context of a user process
- No need for Process Switch to run OS routine



(b) OS functions execute within user processes

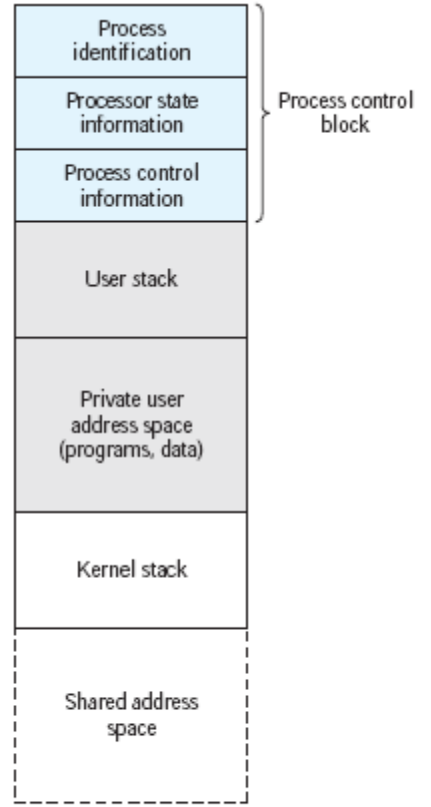


Figure 3.16 Process Image: Operating System Executes within User Space

Process-based Operating System

Process-based operating system

Implement the OS as a collection of system process



(c) OS functions execute as separate processes

Security Issues

An OS associates a set of privileges with each process.

- Highest level being administrator, supervisor, or root, access.

A key security issue in the design of any OS is to prevent anything (user or process) from gaining unauthorized privileges on the system

- Especially – from gaining root access.

System access threats

- Intruders
  - Masquerader (outsider)
  - Misfeasor (insider)
  - Clandestine user (outside or insider)
- Malicious software (malware)

Countermeasures: Intrusion Detection

Intrusion detection systems are typically designed to detect human intruder and malicious software behaviour.
May be host or network based
Intrusion detection systems (IDS) typically comprise

- Sensors
- Analyzers
- User Interface

Countermeasures: Authentication

Two Stages:

- Identification
- Verification

Four Factors:

- Something the individual ***knows***
- Something the individual ***possesses***
- Something the individual ***is*** (static biometrics)
- Something the individual ***does*** (dynamic biometrics)

Countermeasures: Access Control

A policy governing access to resources

A security administrator maintains an authorization database

- The access control function consults this to determine whether to grant access.

An auditing function monitors and keeps a record of user accesses to system resources.

Countermeasures: Firewalls

Traditionally, a firewall is a dedicated computer that:

- interfaces with computers outside a network
- has special security precautions built into it to protect sensitive files on computers within the network.